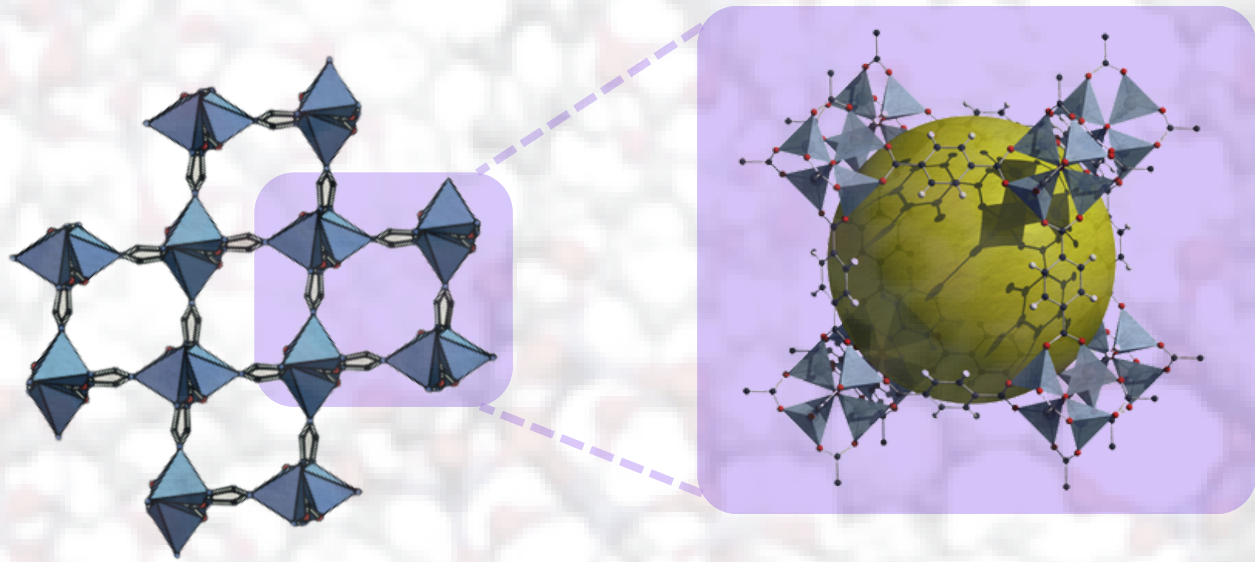


METAL ORGANIC FRAMEWORKS

GROUP 10A

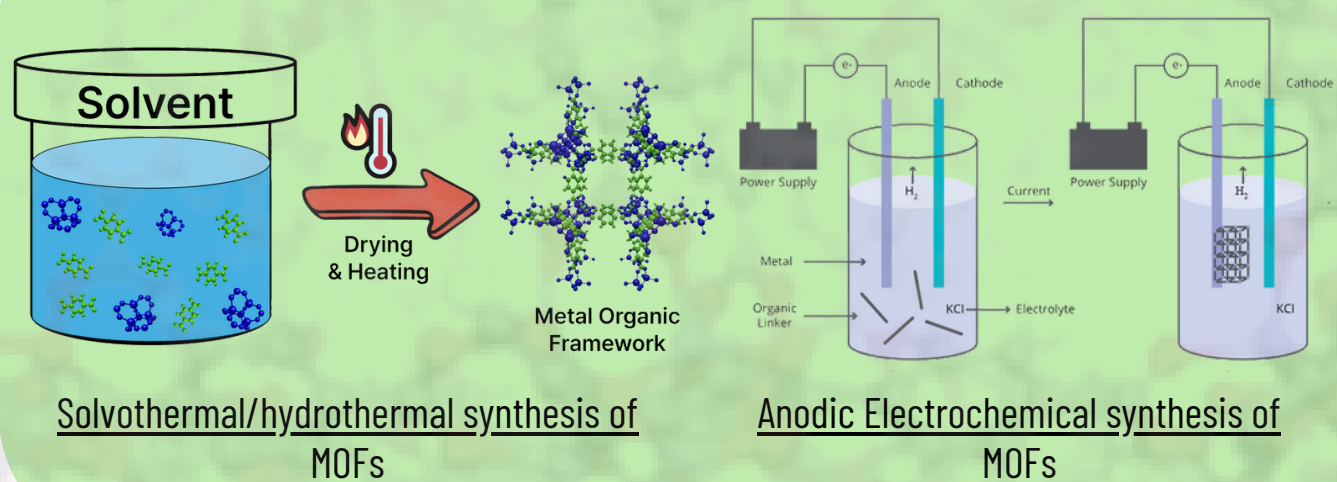
JONGHWI KIM, SEAH YUXIN, TOH KAI XUEN, XU JINGYI



WHAT ARE METAL ORGANIC FRAMEWORKS?

Metal-organic frameworks are complex structures made of metal ions and organic ligands, linkers to create a coordination structure. Due to their large cavities, metal organic frameworks are used for adsorption of several gases, such as carbon dioxide and nitrogen, methane.

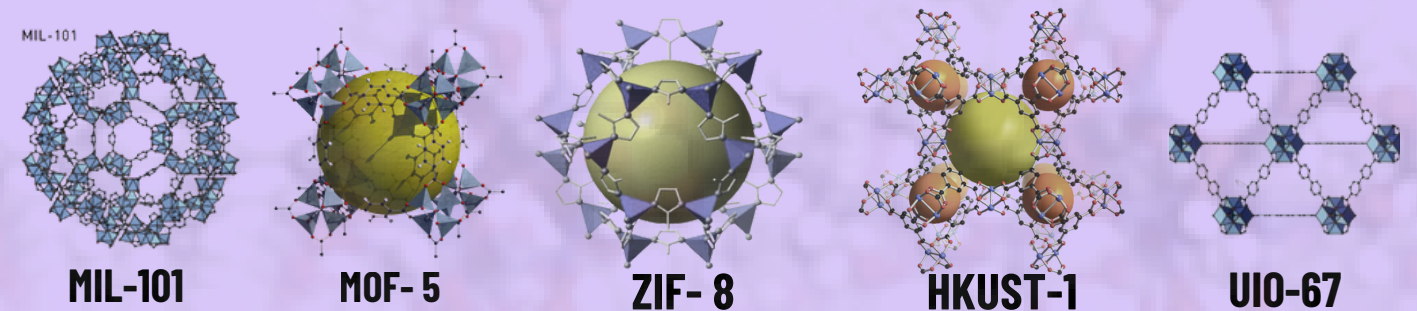
SYNTHESIS OF MOFs



Nobel Prize in Chemistry 2025 - MOF Designing Empty Space

Crystals built from metal nodes and organic linkers that create programmable pores—a new architecture for trapping, separating, and storing molecules.

Types of MOFs

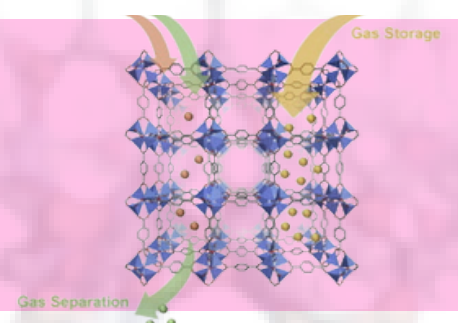


HOW MOFs ARE USED

Metal organic frameworks can be used in many purposes, which include biosensors, batteries, separation sensors, gas storage and many more.

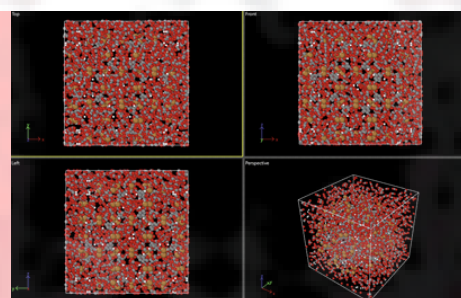
Gas storage

Excellent porosity and surface-to-volume ratio makes MOFs an ideal gas storage, especially because their structural stability before/after storage.



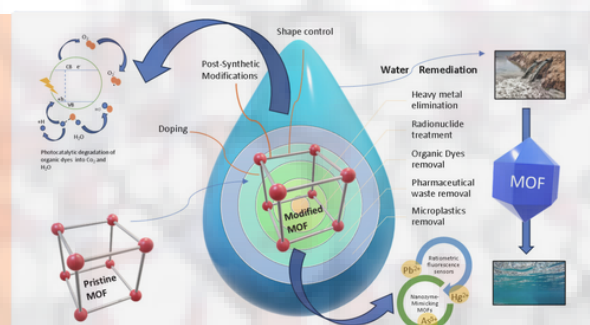
Carbon Capture

Excellent porosity and surface-to-volume ratio makes MOFs an ideal gas storage, especially because their structural stability before/after storage.



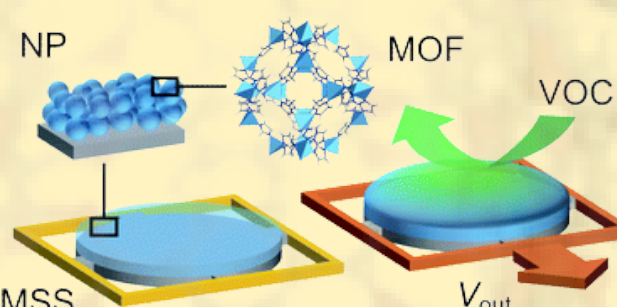
Water purification

Metal organic frameworks can capture and detect many harmful components in water, such as heavy metals, organic dyes and microplastics.



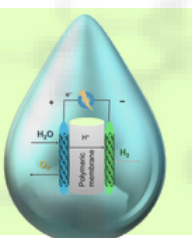
Chemical and Gas Sensing

MOFs can selectively adsorb target molecules into their well-defined pores. These changes may appear as variations in fluorescence intensity, electrical conductivity, or mass.



Hydrogen generation

By performing as a membrane in an electrolytic cell in the electrolysis of seawater, metal organic frameworks can help generate hydrogen.



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PROPERTIES

1. Extremely high surface area to volume ratio, >1000 m²/g
2. Pore size and shape can be precisely controlled and tuned
3. Exhibit selective adsorption toward specific gases or molecules due to their controlled pore environments.
4. Generally lightweight due to their high porosity.

